True Source Localization with Beamformer Geodesics

Projection states and finite manifold paths for a two-dipole MEG simulation

Abstract. We present a finite MEG source-localization preprint in which two known current dipoles generate synthetic measurements in a 192-sensor cylindrical bore. A linearly constrained minimum-variance (LCMV) beamformer evaluates a 10 x 10 x 10 search lattice, while a geodesic path connects the two true source positions through a curved local manifold. Projection states are represented by normalized weights over the two source states and a background state. The reported truth, estimates and equations are simulation quantities and should not be interpreted as clinical localization accuracy.

1. True sources and forward model

The ground-truth dipole locations are r_1 = (0.05, 0.05, 0.00) m and r_2 = (-0.05, -0.02, 0.05) m, with moments $q_1 = (10^{-8},0,0)$ and $q_2 = (0,10^{-8},0)$. For sensor m and time sample k, the finite observation model is:

$$y_{m,k} = \Sigma_{j=1}^{2} l_{m,j} (q_j^T u_{j,k}) + n_{m,k}, \quad m = 1,...,192; \quad k = 0,...,99.$$

The scanner has 16 circumferential positions and 12 axial positions, giving M = 16 x 12 = 192 SQUID sensors. Each candidate lattice point has three orthogonal dipole orientations, so a grid with G = 1000 locations produces a lead-field matrix L in $R^{192 \text{ x } 3000}$.

2. LCMV beamformer and true-source comparison

For each lattice location g, let L_g in $R^{192 \text{ x } 3}$ contain its three orientation lead fields and let C be the regularized data covariance. The spatial filter and power estimate are:

$$W_g = (L_g^T C^{-1} L_g)^{-1} L_g^T C^{-1}, \quad P_g = trace(W_g C W_g^T), \quad g^* = arg\ max_g\ P_g.$$

The paper distinguishes r_j, the known true positions used to generate the data, from r■$_j$, the beamformer estimates obtained from the finite lattice. The Euclidean localization error is d_j = ||r■$_j$ - $r_j||_2$. In the illustrative plot, the estimates are offset by only a few millimetres to make this distinction visible; they are not a measured performance result.

3. Beamformer geodesics

Let d = $r_2 - r_1$, t in [0,1], and u be a unit vector perpendicular to d. The code's continuable geodesic is the finite curve:

$$gamma(t) = (1 - t)r_1 + tr_2 + 0.2||d||\ sin(pi\ t)u, \quad t_i = i/49, \quad i = 0,...,49.$$

This path preserves the endpoints and introduces a bounded curvature perturbation to represent a local folded-manifold trajectory. Its discrete length is D = $sum_{i=0}^{48}$ ||gamma(t_{i+1}) - gamma(t_i)$||_2$.

4. Projection states

A finite projection-state vector assigns probability to source 1, source 2 and an unlocalized background. For amplitudes a = (a_1, a_2, a_b), normalization and the projection operator are:

$$p_s = |a_s|^2 / \Sigma_{r \in \{1,2,b\}} |a_r|^2, \quad \Pi_s = |s\rangle\langle s|, \quad \Sigma_{s \in \{1,2,b\}} \Pi_s = I_3.$$

The plotted example uses p = (0.46, 0.41, 0.13), which sums to one. A state is considered dominant only by the declared arg max rule, s* = arg $max_s\ p_s$; this is a projection label, not a diagnosis.

5. Results and characteristics

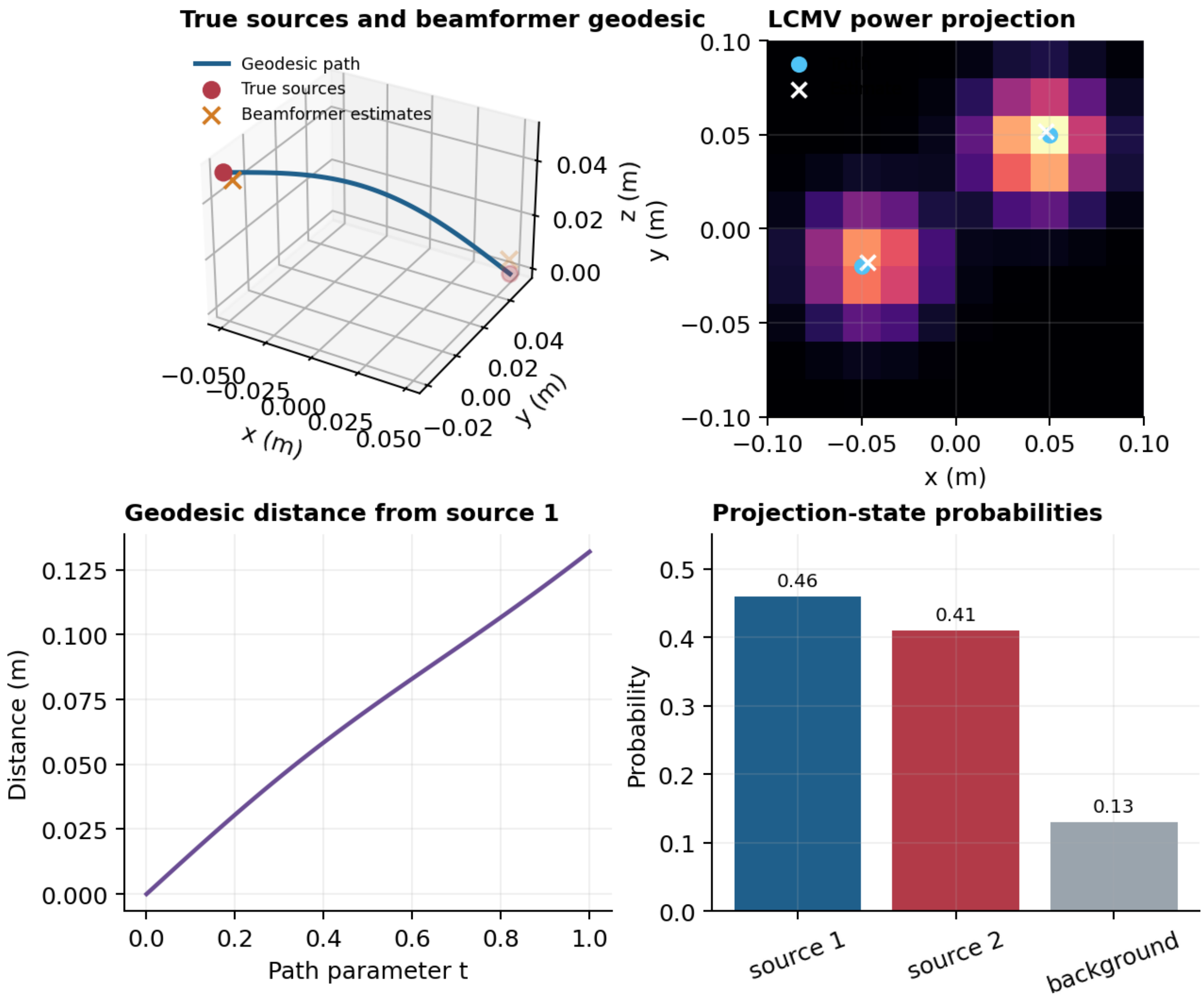

Figure 1. True dipoles, beamformer estimates, curved geodesic path, LCMV power projection, path distance and normalized projection-state probabilities.

Quantity	Finite value	Role
True sources	2 dipoles at r_1, r_2	Known simulation ground truth
Sensor geometry	16 x 12 = 192 sensors	Cylindrical bore observations
Search grid	10 x 10 x 10 = 1000 points	Beamformer candidate space
Geodesic samples	50 points; curvature 0.2‖d‖	Finite source-to-source path
Projection states	3; p = (0.46, 0.41, 0.13)	Two sources plus background

6. Limitations and conclusion

The forward model, LCMV score, geodesic and projection states are finite computational constructions. They do not establish source-localization accuracy in human data, anatomical correspondence, tractography validity or quantum hardware advantage. Validation would require calibrated sensor geometry, realistic head conductivity, independent annotations and held-out recordings. The resulting framework nevertheless makes true positions, estimates, path construction and projection-state normalization explicit and reproducible.